

# OmniView Architecture Standard Whitepaper

## Section 1.0 - Project Overview & Evolution (100% MIT License)

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Visual Studio Code has become the primary workbench for global developers.

OmniView delivers instant, lightweight, zero-paywall previewing for Markdown, Mermaid, PlantUML, SVG, and real

Designed with a decoupled driver micro-kernel to keep RAM overhead under 45MB while eliminating commercial V

### OmniView Multi-Driver Highlights

- 100% Free & Open-Source under MIT License
- Multi-core Canvas rasterization with Web Worker acceleration
- Zero telemetry, fully offline & zero network dependencies

# Sandboxed IPC Protocol & Security Architecture

## Section 2.0 - Security & Protocol Specification

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The Extension Host (Node.js) and Webview (Chromium) operate in isolated process boundaries. All communications flow over typed JSON-RPC message contracts with bidirectional live hot reload. Rigorous Content Security Policies (CSP) and DOMPurify sanitization strictly eliminate XSS attack vectors.

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# Multi-Driver Lazy Loading Architecture

## Section 3.0 - Driver Architecture Matrix

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Each format (Markdown, PlantUML, SVG, CSV, PDF, Typst, Excalidraw, Notebook) is encapsulated into an isolated driver. The PDF Driver utilizes Mozilla PDF.js v4+ with Web Worker acceleration and Canvas rasterization. Provides full resolution Hi-DPI multi-page navigation, smooth zooming, and clockwise 90 degree rotation.

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# Deployment & Verification Runbook

## Section 4.0 - Production Readiness Checklist

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Packaged into standard .vsix extensions using official @vscode/vsce tooling.

Continuous integration pipeline validates zero memory leaks and 100% diagram diagnostic test passes.

100% Free & Open Source under the permissive MIT license for the entire developer community.

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